



School Of Engineering

Image Processing - ENGG*6302: Winter 2025

Assignment 1

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Question 1

For this question, a 2D MRI image of the liver was chosen. First, the image is displayed using the `imshow()` function from the `matplotlib` python library (Figure 1). Then, in the second window, a corresponding grayscale histogram of the image is displayed (Figure 2). The histogram displays the distribution of pixel intensity values, with a noticeable peak at lower intensity values, which primarily corresponds to the background.

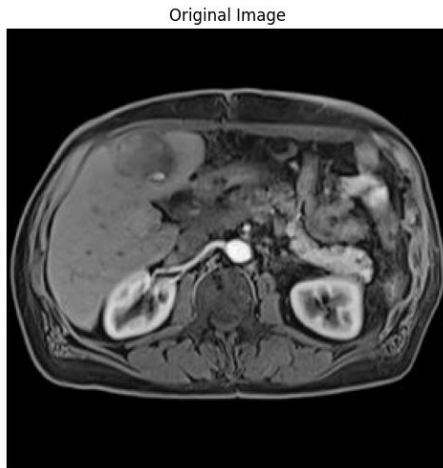


Figure 1. Original image of a 2D Liver

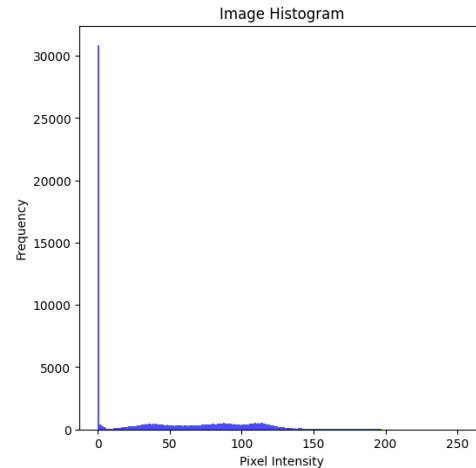


Figure 2. Histogram of the 2D grayscale liver image

The selection of specific rows and columns for this image analysis involved a process of trial and error to identify areas that best capture both homogeneous regions and distinct edges. The interesting row and column chosen for this image was row 40 and column 5 (Figures 3 and 4). This is because they capture both homogeneous regions and edges in the image. Figure 3 displays fluctuations in pixel intensity with two distinct peaks. These peaks likely correspond to boundaries between different regions in the image, capturing areas with significant contrast changes. The broad flat region at the lower end of the graph represents homogenous areas, while the sharp spikes signify transitions, possibly between tissue structures or edges within the liver. The intensity profile is useful for understanding how the image's brightness varies across this specific row.

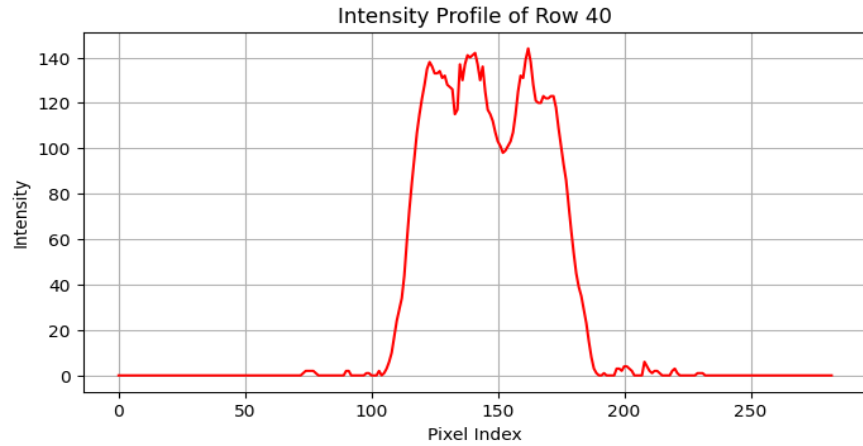


Figure 3. Intensity profile of row 40

In Figure 4, the intensity profile for column 5 is shown. Like row 40, this plot reveals fluctuations in intensity, but with a slightly different pattern. There are two noticeable peaks, which suggest distinct regions with sharp intensity transitions, indicating boundaries or edges. The areas of low intensity at the base correspond to more uniform or smooth regions. The intensity profile of column 5, when compared with row 40, may highlight vertical boundaries between structures in the liver, adding further detail to the spatial distribution of intensity values in the image.

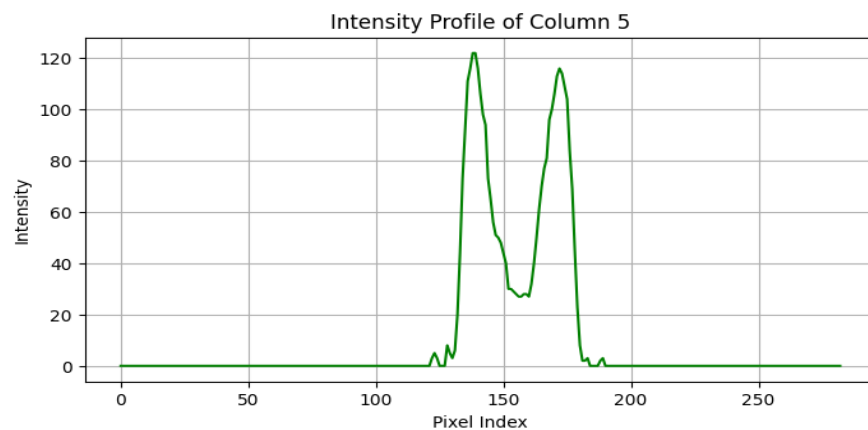


Figure 4. Intensity profile of column 5

Figure 5 shows the kernel density estimation (KDE) of image intensity, showing the estimated probability distribution for pixel intensity values. The blue curve represents the density for a bandwidth of 0.8, while the red dashed curve represents a bandwidth of 1.8. The KDE is useful for understanding the overall distribution of intensity values across the image, highlighting areas with concentrated intensities and those with more variation. The difference in bandwidths provides insight into how smoothing affects the density estimation. A smaller bandwidth (blue) captures more detailed intensity fluctuations, while a larger bandwidth (red) produces a smoother, broader distribution, reflecting the overall characteristics of the image's intensity across the entire dataset.

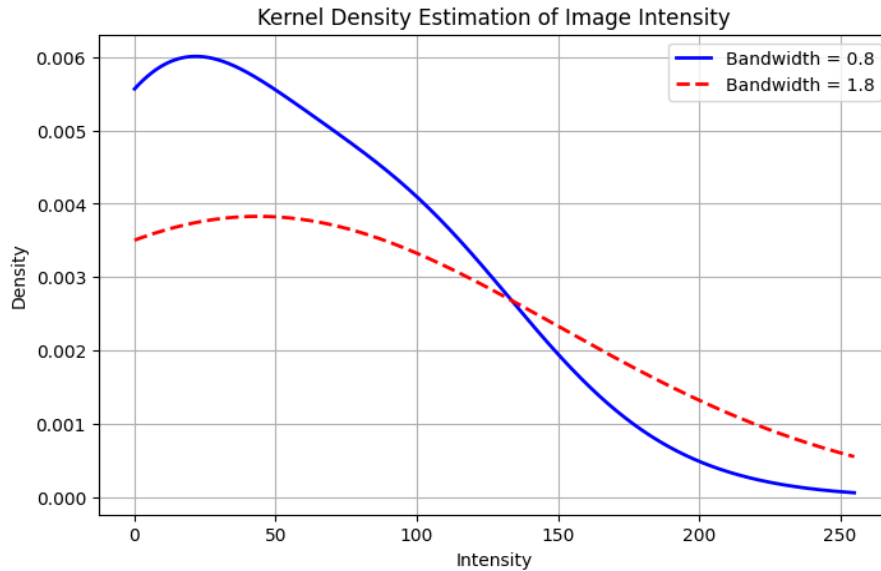


Figure 5. Kernel density estimation of image intensity for bandwidth of 0.8 (blue) and bandwidth of 1.8 (red)

Question: How many modes does the histogram have? Can you identify each mode with specific structures in the image? Explain. [100 to 150 words]

The histogram of the image

The histogram of the image shows multiple modes, corresponding to different intensity regions within the image. If the image contains distinct structures such as organs, tissues and background, the histogram will likely have peaks representing these regions. For example, a grayscale medical image of the liver may show one peak for dark background areas, another for soft tissues and a third for brighter regions such as blood vessels or lesions. Each mode in the histogram suggests a different structural component in the image. By analyzing the intensity distribution, we can infer how different parts of the image contribute to the overall contrast and segmentation.⁹

Question: How many modes are in the PDFs? Compare it to the modes you saw in the histogram of the same image.

When computing the PDF using KDE, the number of modes depends on the bandwidth. A smaller bandwidth (e.g. 0.01) captures more details, potentially revealing more modes, while a larger bandwidth (e.g., 1.8) smooths the PDF, reducing the number of visible modes. Comparing the KDE modes to the histogram, the KDE might show similar or additional modes, especially with smaller bandwidths, as it can detect subtle intensity variations. However, with larger bandwidths, the KDE might merge modes, making it less detailed than the histogram. The choice of bandwidth thus affects how well the PDF represents the underlying intensity distribution of the image.

Question 2

The 2D greyscale lung MRI scan was chosen for this question. In this example, we have to apply 3x3 average and median filter on a medical image and also analyze the impact on an extracted row and

column. The average filter averages each pixel with its neighbors, reducing noise but smoothing features, causing blurriness, as shown in Figure 6.

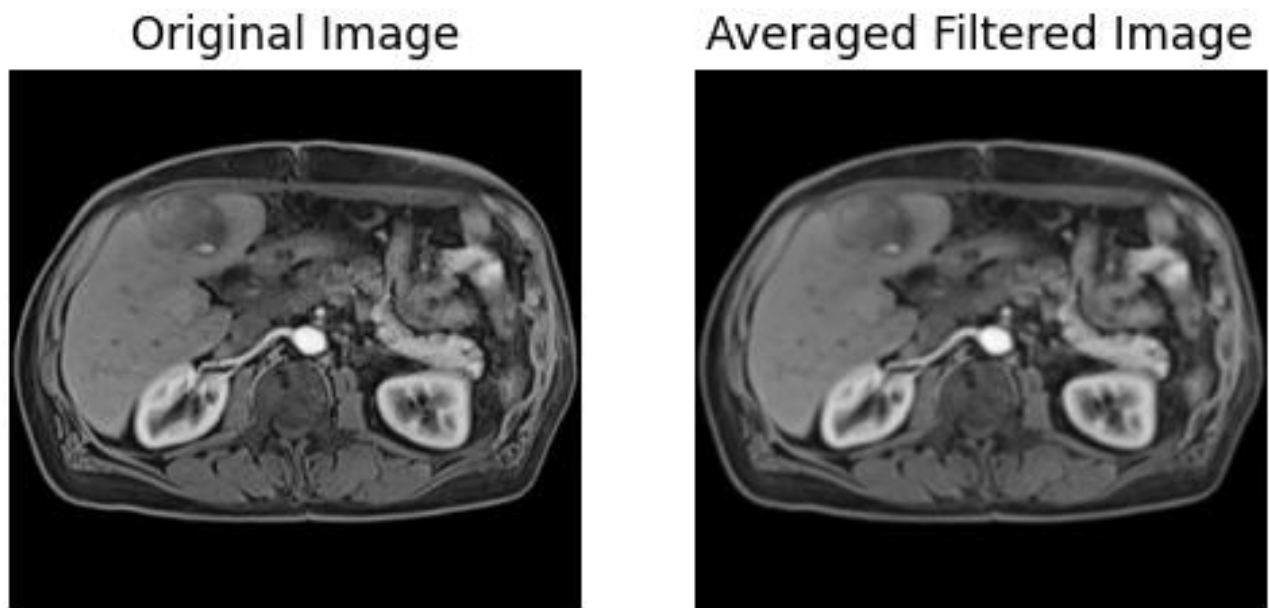


Figure 6. Comparison of original image (left) vs. average filtered image (right).

However, median filter replaces the pixel with median value of its neighboring pixel that reduces noise while preserving more feature than the average feature for which the image has a reduced blurring image as shown in Figure 7.

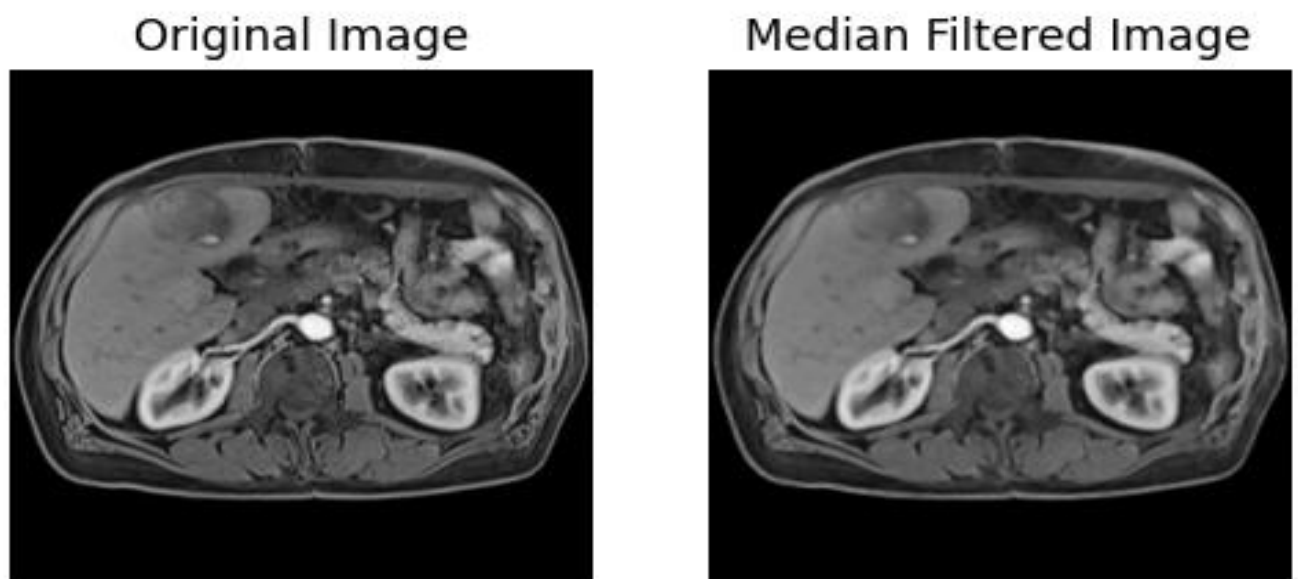


Figure 7. Comparison of original image (left) vs. median filtered image (right).

In the next step, a specific row (40) and column (5) was extracted from the image to analyze the impact of an image. The smoothing effect is evident in Figure 8 and Figure 9 with both filters reducing noise while the median filter captures the variation more effectively.

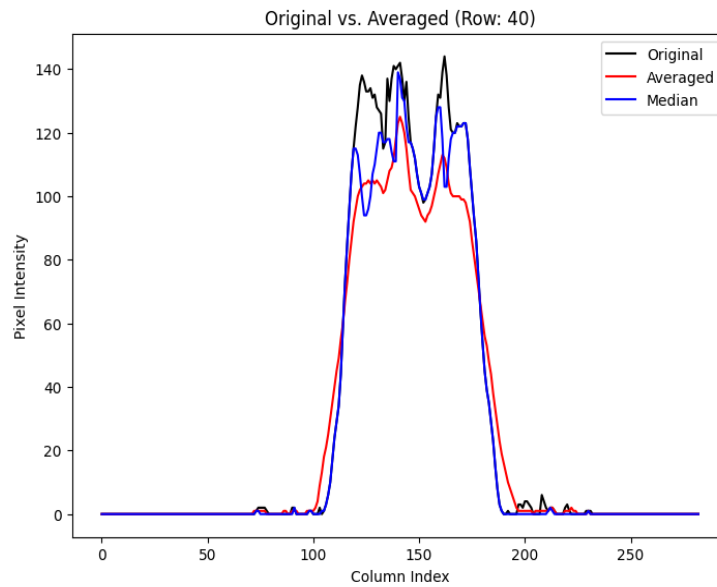


Figure 8: Pixel Intensity for Row 40 Original vs. Filtered (Averaged and Median).

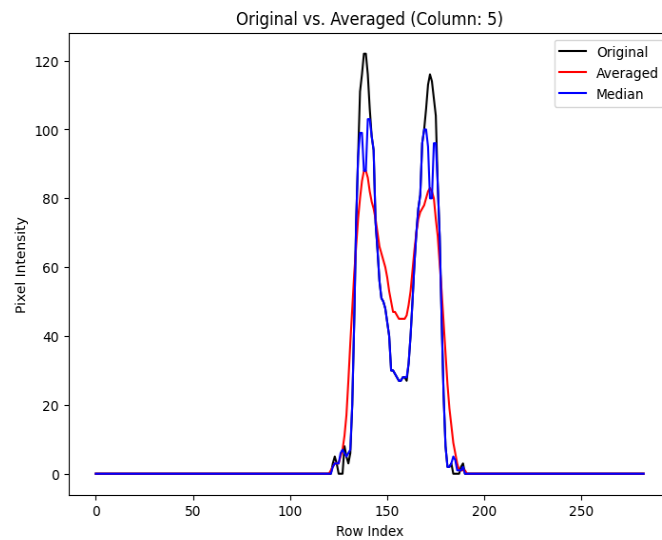


Figure 9. Pixel Intensity for Column 5 Original vs. Filtered (Averaged and Median)

Question: Based on both the image and the extracted row or column, compare the original image, the average-filtered image and the median-filtered image. What happens to homogeneous areas of the image? What happens to noisy areas of the image? What happens to edges? [100 to 150 words]

In Figure 8 and Figure 9, the left and right parts of the row and column intensity levels represent homogeneous regions (no sharp changes) where both filters preserve almost the original intensity by averaging or replacing pixel values within their neighborhoods. In the middle, sharp changes indicate non-

homogeneous areas and transitions that mark edges. The average filter smooths these changes by computing mean values, which blurs edges and reduces detail, while the median filter, by selecting the median value, preserves transitions and sharp features more effectively. We can observe that both filters smooth areas with sharp changes, demonstrating their ability to reduce outliers or noise, particularly in non-homogeneous regions.

Question 3

Salt and Pepper (S&P) noise randomly replace some pixels with either minimum & maximum intensity values (black and white). This creates a distinct random pattern of noise.

Gaussian noise follows a normal distribution and affects all pixel values to some extent, leading to a more evenly spread distortion like a blur distortion.

For S&P noise the probability of a pixel being corrupted determines the noise intensity, in this case a $p=0.02$ was chosen. Higher probabilities introduce more speckling, making it harder to reconstruct the original image. A bigger image was used for test and the noise was not as visible as in this smaller image.

Gaussian noise is defined by the mean and standard deviation (STD) in which a larger STD increases the spread of pixel intensity variations, creating a smooth but distorted effect resemblant of blur. In this case, a configuration of mean=0 and standard deviation (std) or 25 was chosen. Results can be seen in **Error! Reference source not found..**

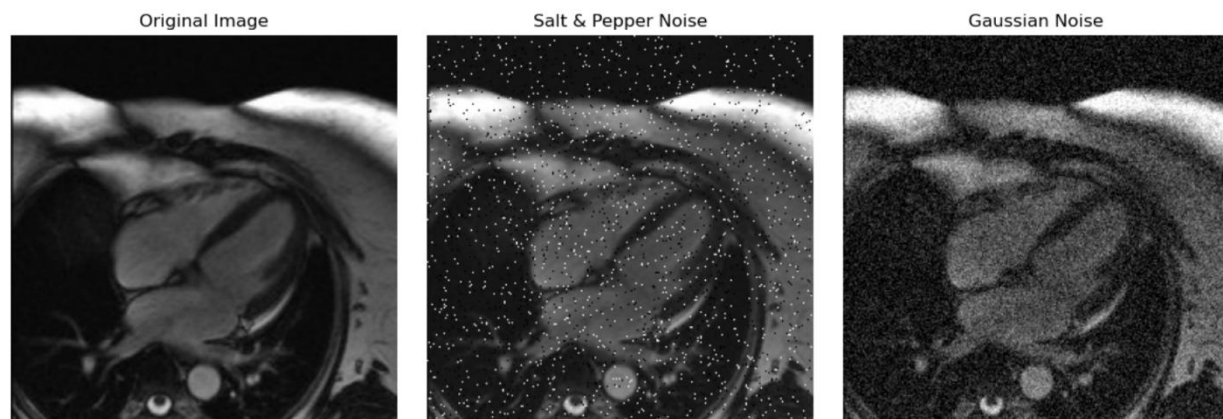


Figure 10. (From left to right) original liver image, image with Salt & Pepper noise added, image with Gaussian noise added

Question: Specify linear (i.e., Gaussian filter) and order statistics filters (i.e. median filter) with different mask sizes, say for example, 3×3 , 9×9 , and 15×15 . Apply both types of filters using the same mask size to the noisy images and compare their performance in terms of mean squared error and general visual properties. Include snapshots and observations in your report.

The Median 3×3 filter provides the most visually appealing results with minimal blurring. For that same mask size, the Mean filter performs acceptably in visual terms, but some noise is still evident in the image. The Gaussian filter offers a balance between noise reduction and edge preservation, but some noise remains visible.

For mask size of 9×9 , the median filter preserves a lot of edges visually and no noise is seen anymore, but some other image characteristics are lost as can be seen in Figure 11. The mean filter, however, introduces excessive blurring which makes edges appear faded. The Gaussian filter smooths the image while preserving better detail than the mean filter but still shows slight blurring.

For a 15x15 mask, the median filter removes nearly all noise but also erases finer textures. The mean and Gaussian filters blur the image heavily leading to loss of sharpness. Overall, the median filter performs best in noise removal while maintaining critical image structures.

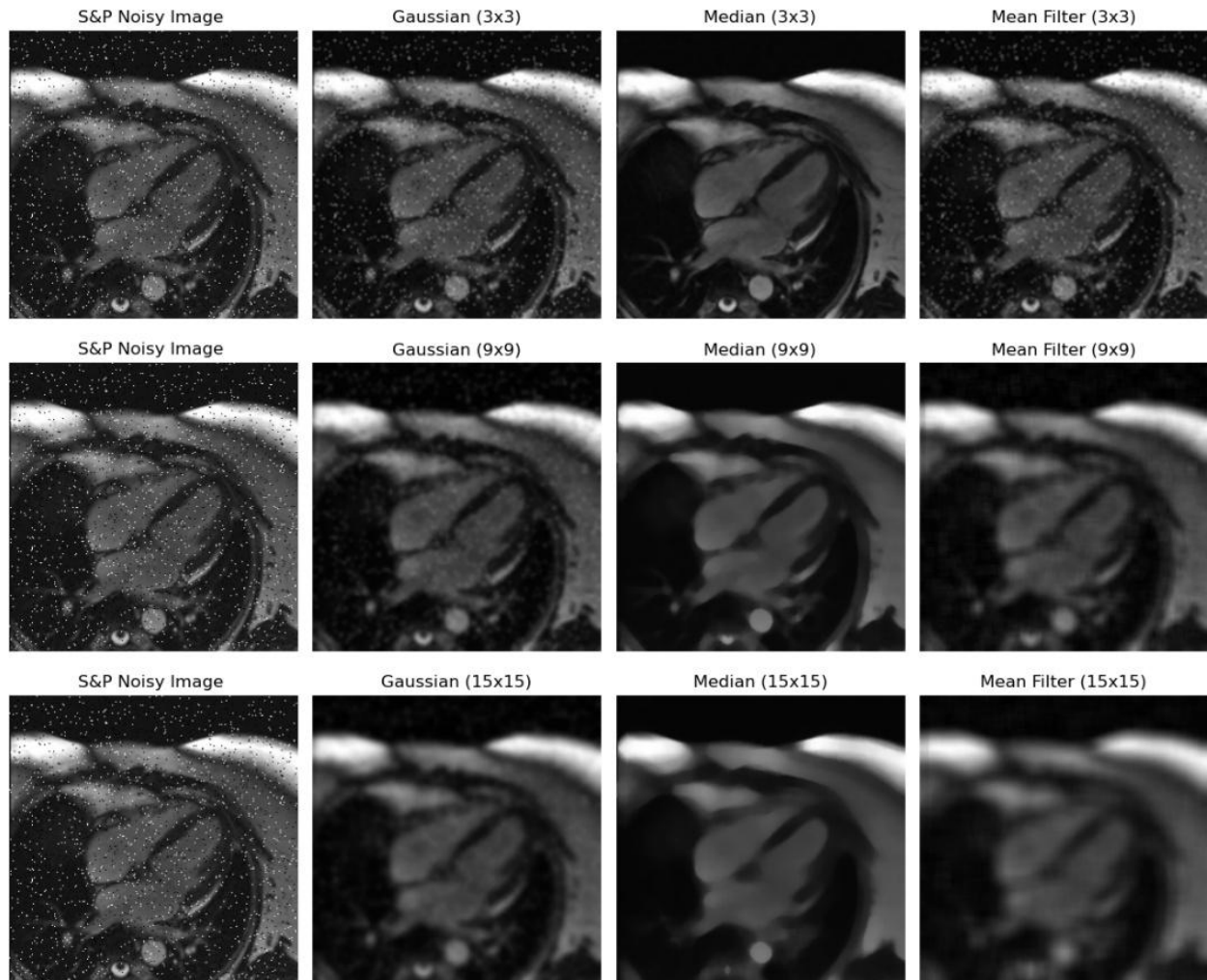


Figure 11. Result of different filter masks on Salt & Pepper noise. S&P noise image (column 1), Gaussian filter (column 2), Median filter (column 3), Mean filter (column 4) .

Below, Table 1 shows the mean squared error (MSE) for the filters and mask sizes applied with the S&P noise image. With these results we confirm that the median 3x3 filter had the best results, followed by median 9x9.

Table 1. MSE for filters applied with Salt & Pepper noise.

Size / Filter	Gaussian	Median	Mean Filter
3x3	144.19	5.70	123.40
9x9	82.17	57.84	136.11
15x15	116.45	146.26	260.83

We then proceed with the Gaussian noise image, applying the same filters and mask sizes as can be seen in Figure 12 below.

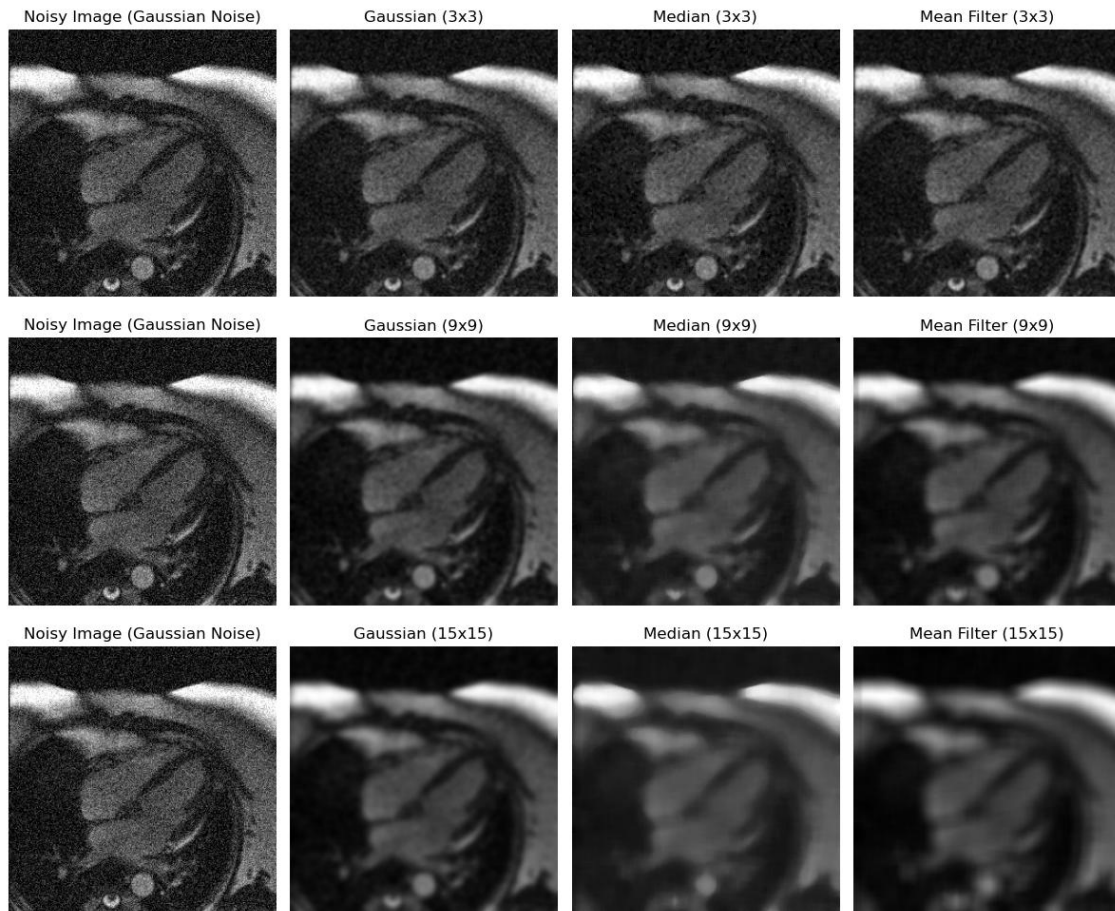


Figure 12. Result of different filter masks on Gaussian noise. S&P noise image (column 1), Gaussian filter (column 2), Median filter (column 3), Mean filter (column 4).

The Gaussian filter produces the smoothest results while preserving important image details, but large mask sizes create a lot of blur in the image. The median filter 3x3 (which performed well with S&P noise), shows poor performance, preserving the noise in the image. The MSE results for the operations can be seen in Table 2 below. The best performing filter was Gaussian 9x9.

Table 2. Mean Squared Error (MSE) for filters applied with Gaussian noise.

Size / Filter	Gaussian	Median	Mean Filter
3x3	83.41	105.79	72.85
9x9	62.30	84.76	121.28
15x15	101.20	171.35	247.50

Larger mask sizes generally increase blurring, visually reducing noise but also representing a loss of detail (visual information of the image). The Mean filter performs acceptably well in both cases but creates significant blurring with larger mask sizes, making it less reliable for applications requiring edge preservation or where image information is prone to be lost.

Question 4

Laplacian-based contrast enhancement is a technique used to sharpen edges and enhance details in an image. In Python, we can achieve this by using the Laplacian of Gaussian (LoG) method, where the standard deviation (σ) of the Gaussian function controls the smoothing and enhancement effects. Smaller σ values (e.g. 0.5) enhance fine details but may amplify noise, whereas larger σ values (e.g. 2.0) smooth the image and highlight broader edges.

Figure 13 presents the original and contrast-enhanced images for different σ values. The top left shows the original image, while the other three images demonstrate enhancement at $\sigma = 0.5$, $\sigma = 1.0$ and $\sigma = 2.0$. As σ increases, edges appear more pronounced but excessive smoothing can reduce finer details.

After applying the LoG filter, the enhanced image is obtained by subtracting the filtered results from the original image. Displaying the original and enhanced images side by side allows us to visually compare the results. Additionally, superimposing a selected row or column from the original and enhanced images helps analyze how pixel intensities change after enhancement.

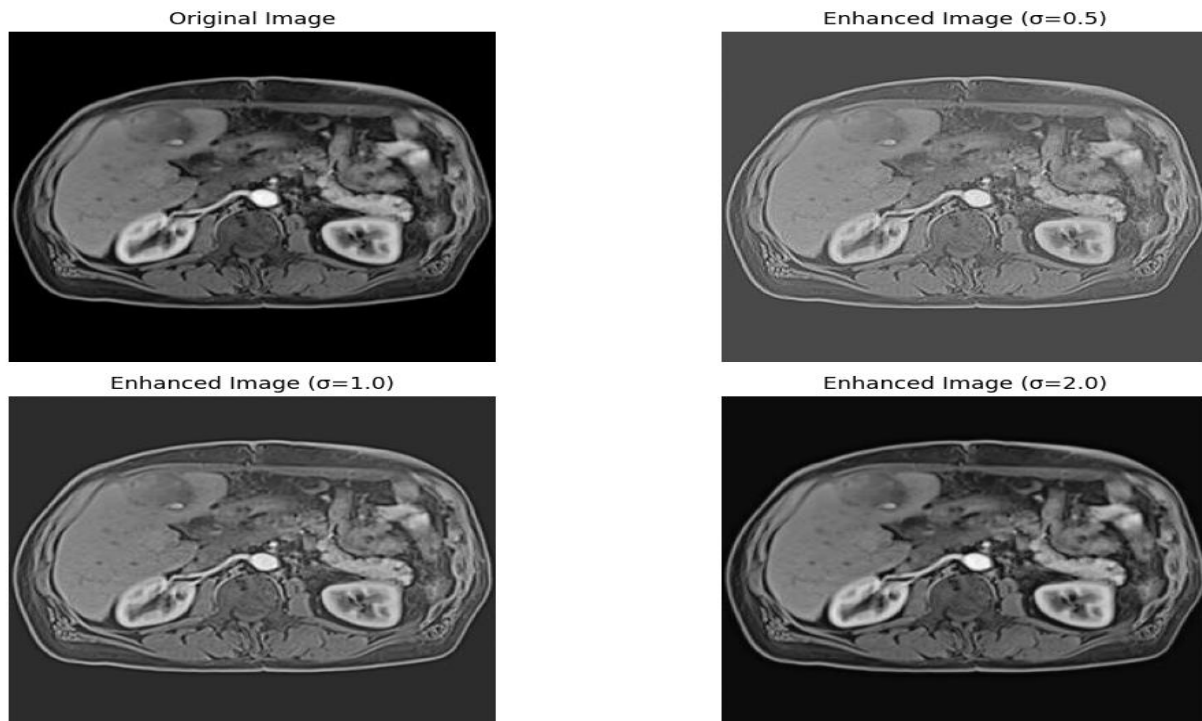


Figure 13. Original image (top left), enhanced image with sigma = 0.5 (top right), enhanced image with sigma = 1 (bottom left) and enhanced image with sigma = 2 (bottom right).

Figure 14 provides a quantitative comparison by plotting the intensity profile of a selected row across different enhancements. The original image (represented by a black line) exhibits higher variability and sharper intensity fluctuations, indicating the presence of noise and fine details. In contrast, the enhanced images show progressively smoother profiles as the sigma value increases, demonstrating the effect of LoG filtering in reducing noise and enhancing contrast.

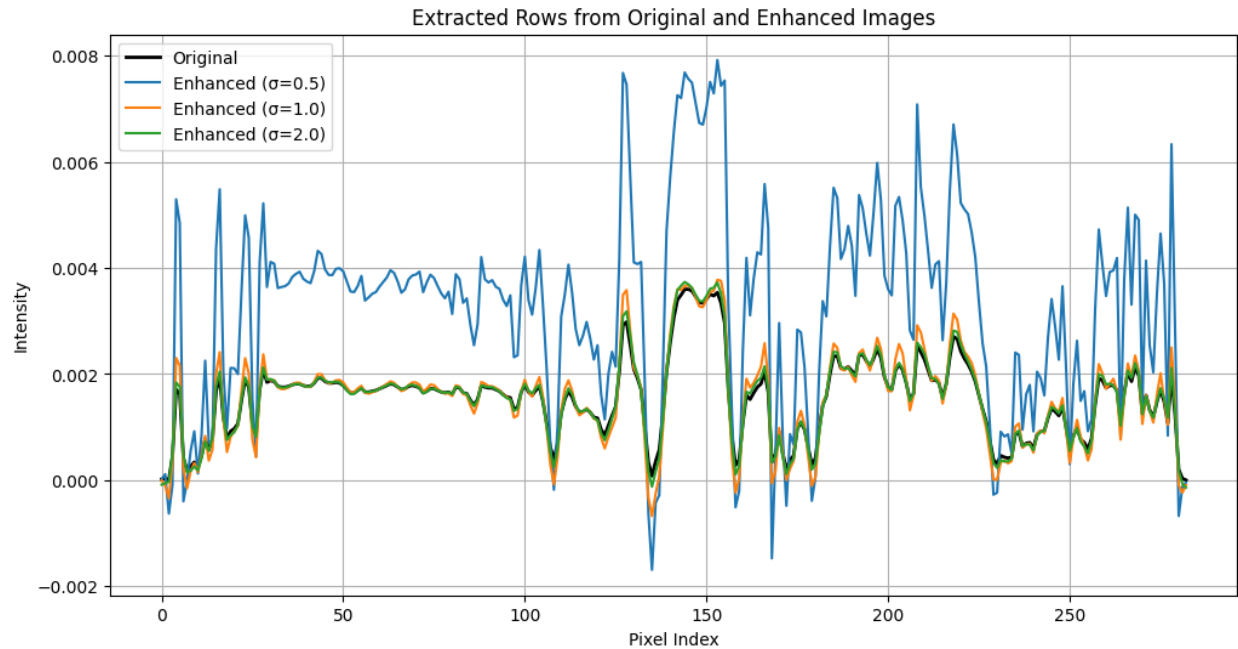


Figure 14. Extracted Rows Profile from the original and enhanced images

Question: Based on both the image and the extracted row or column, compare the original image and the contrast-enhanced images. Which value of alpha provides the best visual results on your image? Explain. What happens to homogeneous areas of the image? What happens to noisy areas of the image? [100 to 150 words]

Based on the image and the extracted row/column, the contrast-enhanced images show improved edge sharpness and detail visibility compared to the original image. The best visual results are achieved with a moderate σ value (e.g., 1), as it effectively enhances edges without overly amplifying noise. Smaller σ values (e.g. 0.5) enhance fine details but introduce noise while larger σ values (e.g. 2.0) smooth the image but may lose some fine details.

In homogeneous areas (smooth regions), the contrast enhancement has minimal effect, as these regions lack significant intensity variations. In noisy areas, smaller σ values amplify noise, making the image appear grainier, while larger σ values reduce noise but may also blur fine details. Thus, the optimal σ balances detail enhancement and noise suppression providing the best visual results.

Question 5

Question: Please explain in your report why you will most likely not observe a uniform distribution after histogram equalization?

When histogram equalization is applied to an image, the desired result is always to get an evenly distributed histogram but, in most cases, we get an optimized histogram instead of an ideal one. In a natural image, we often get clusters of intensities because of pixels from an underlying structure, corresponding to different objects. Histogram equalization will redistribute intensities but cannot create a perfect uniform

histogram due to these dominant intensity clusters. However, it's more challenging to get an evenly distributed equalized histogram for a medical image. In this exercise, histogram equalization was applied to a medical image and the result is shown in Figure 15.

In Figure 15, we can observe a sharp peak at lower intensity values in the original histogram, indicating many dark pixels. The histogram equalized image shows improved contrast, making different anatomical structures more distinguishable. However, if we analyze the equalized histogram, it is more spread out compared to the original but still retains noticeable peaks and is not perfectly uniform. This happens because the dominant intensity clusters from the original image are not completely redistributed. The black background remains prominent even after equalization, leading to a high peak at low intensities. Moreover, there is also an effect of nature of digital image and quantization effects, where rounding during intensity mapping creates gaps and spikes instead of a smooth distribution. From the above discussion, it's evident that for medical images, histogram equalization improves contrast but does not produce perfectly uniform histogram. The best possible outcome is a more spread-out distribution, but true uniformity is not achievable due to image structures and digital quantization.

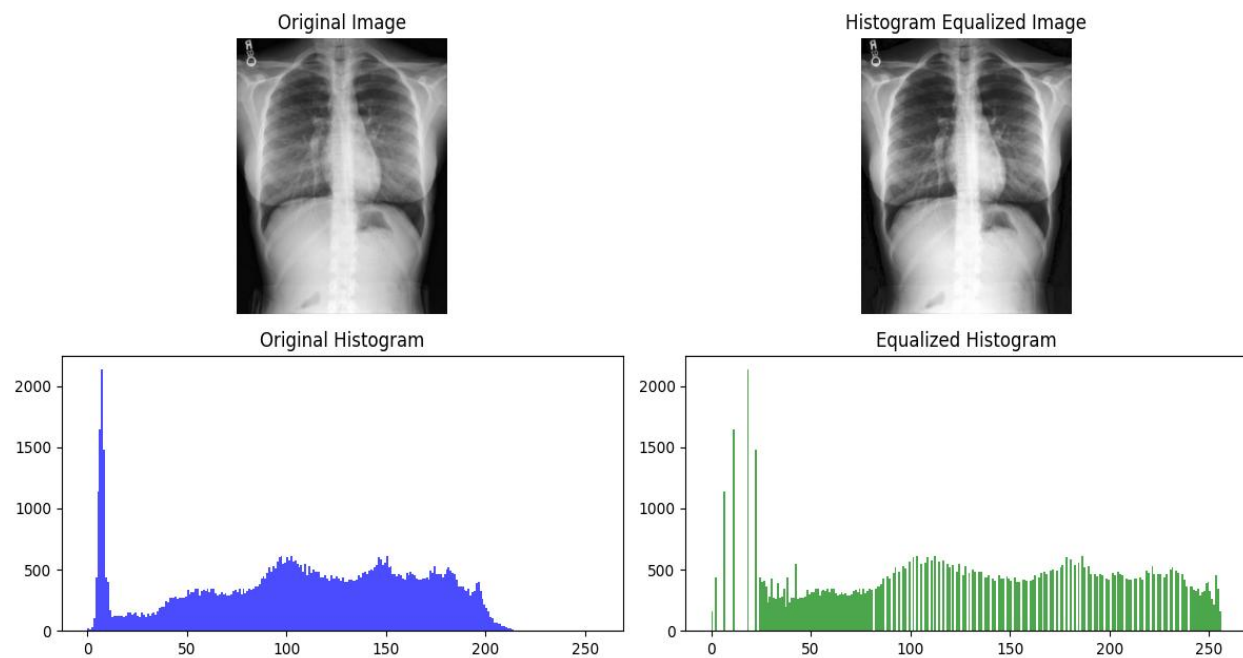


Figure 15. Original X-Ray image (left) vs. histogram equalized Image (right) with their histograms

Question 6

To perform filtering in the frequency domain, we first must apply the 2D Fourier Transform to the medical image. We chose to go with the liver image. Figure 16 shows the original liver image on the left and the frequency spectrum of the image on the right.

The Fourier Transform was used to convert the image from the spatial domain (pixel intensities) to the frequency domain, where the image is represented in terms of its frequency components. Low-frequency components correspond to smooth regions and gradual intensity changes, while high-frequency components

correspond to edges, fine details and noise. After computing the Fourier Transform, we shift the zero-frequency component to the center of the spectrum for easier visualization and filtering.

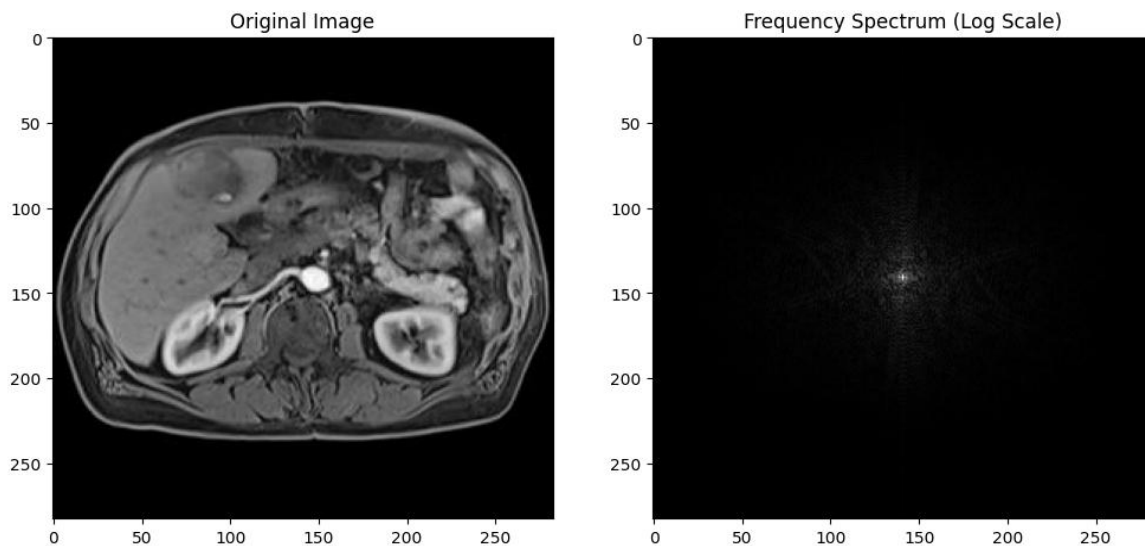


Figure 16. Original image (left) and image in frequency domain using log scale (right)

For low pass filtering, we create a circular mask centered on the frequency spectrum. This mask allows low-frequency components (near the center) to pass while attenuating high frequency components (toward the edges). The filtered frequency domain image is then converted back to the spatial domain using the inverse Fourier Transform. The resulting image appears smoother, with reduced noise and less prominent edges as can be seen in Figure 17.

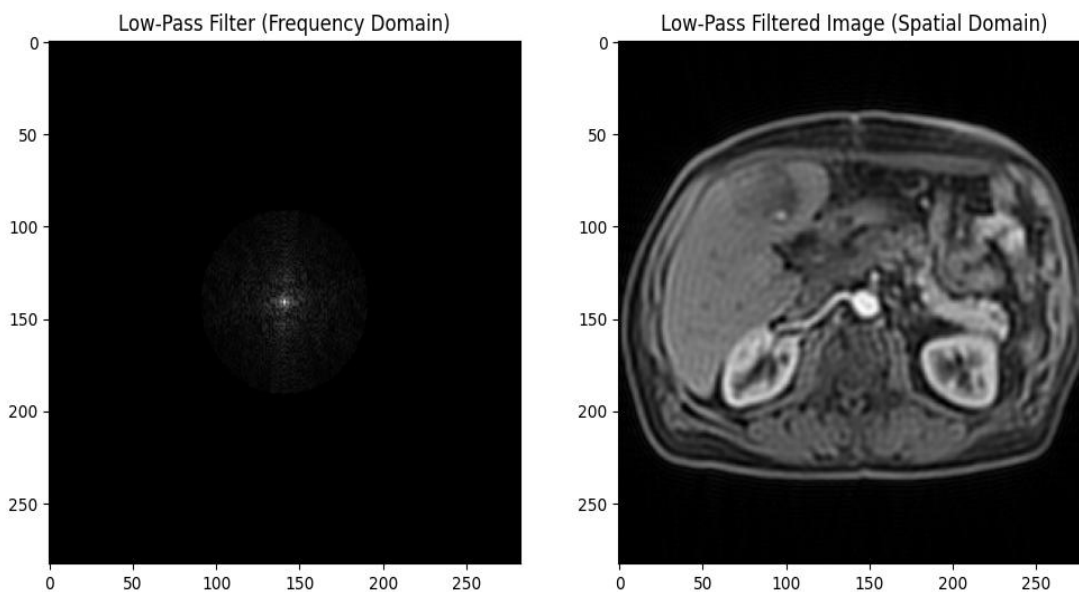


Figure 17. Low-pass filter in frequency domain (left) and low-pass filtered image in spatial domain (right)

For high pass filtering, we used the inverse of the low pass mask, which allows high-frequency components to pass while attenuating low frequency components. The high pass filtered image retains edges and fine details but loses smooth regions as can be seen in Figure 18.

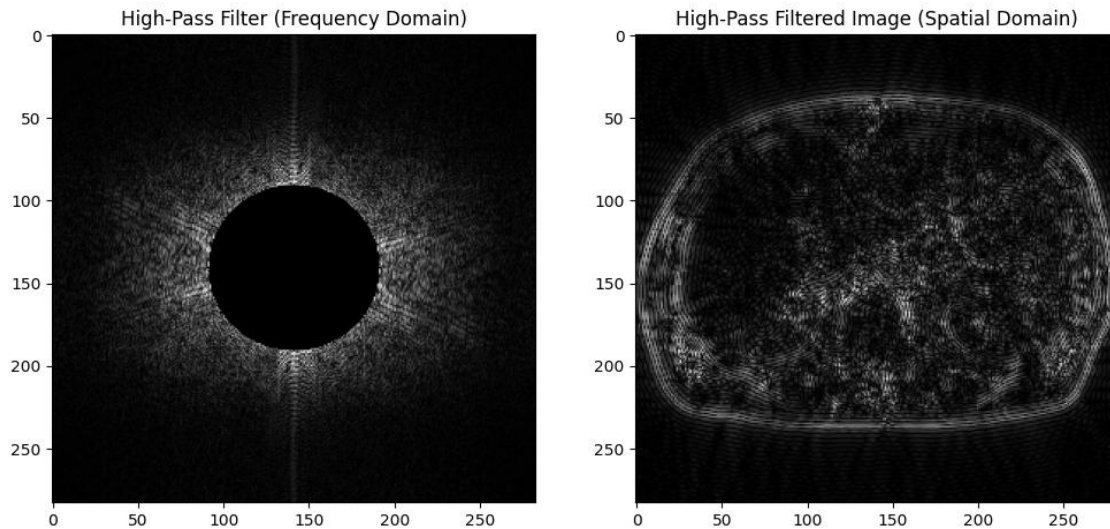


Figure 18. High-pass filter in frequency domain (left) and high-pass filtered image in spatial domain (right)

Question: What is the advantage of performing filtering in the frequency domain (as opposed to performing it in the spatial domain)?

There are several advantages of performing filtering in the frequency domain over the spatial domain. First, filtering in the frequency is computationally more efficient, especially for larger images because convolution in the spatial domain is equivalent to simple multiplication in the frequency domain. Second, frequency domain filters are easier to design and visualize. For example, low pass and high pass filters can be created by masking specific regions of the frequency spectrum, which is more intuitive than designing spatial domain kernels. Third, frequency domain filtering allows precise separation of frequency components, which makes it easier to isolate and enhance or suppress specific features, such as edges or noise. Finally, frequency domain operations affect the entire image globally, ensuring uniform enhancement or suppression of features across the image. In contrast, spatial domain filters operate locally, which can lead to uneven results. Overall, frequency domain filtering provides a powerful and efficient way to process images, particularly for tasks like noise reduction, edge enhancement and feature extraction.

Question 7

The medical image chosen for this question is the 2D grey level MRI liver image shown in Figure 19.

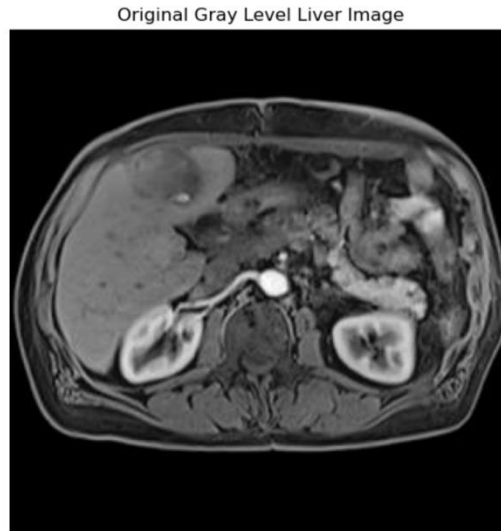


Figure 19. Original grey medical image.

Question: Compare and discuss these edge detection methods. [100 to 150 words]

Edge detection methods identify intensity transitions in an image, with the choice of method depending on the image's characteristics. Sobel and Prewitt filters highlight edges by computing gradients in horizontal and vertical directions, but Sobel gives more weight to central pixels, making it more effective in noisy images. Canny Edge Detection is a multi-step process that smooths the image, computes gradients, and tracks edges via hysteresis. It provides refined edge detection and is less affected by noise. The Laplacian of Gaussian (LoG) method applies Gaussian smoothing before computing the Laplacian, enabling edge detection in multiple directions. While effective for detecting detailed structures, it is highly sensitive to noise. Figure 20 shows the results of applying kernels of sizes 3, 7, and 15. Larger kernels caused edge bleeding to increase exponentially, but no major differences were observed aside from this effect.

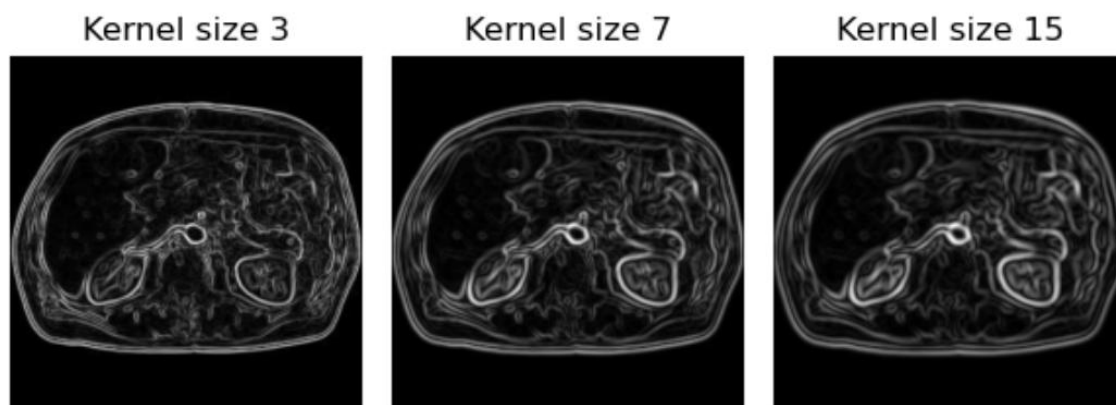


Figure 20. Results of different kernel sizes applied to the image, kernel size 3 (left), kernel size 7 (center) and kernel size 15 (right)

Following the same logic, different edge map percentiles were chosen for which the results can be observed in Figure 21. Percentile 55 marked edges where they did not exist in the original image, and 95th percentile lost a lot of existing edges. The 75th percentile edge map showed the best results.

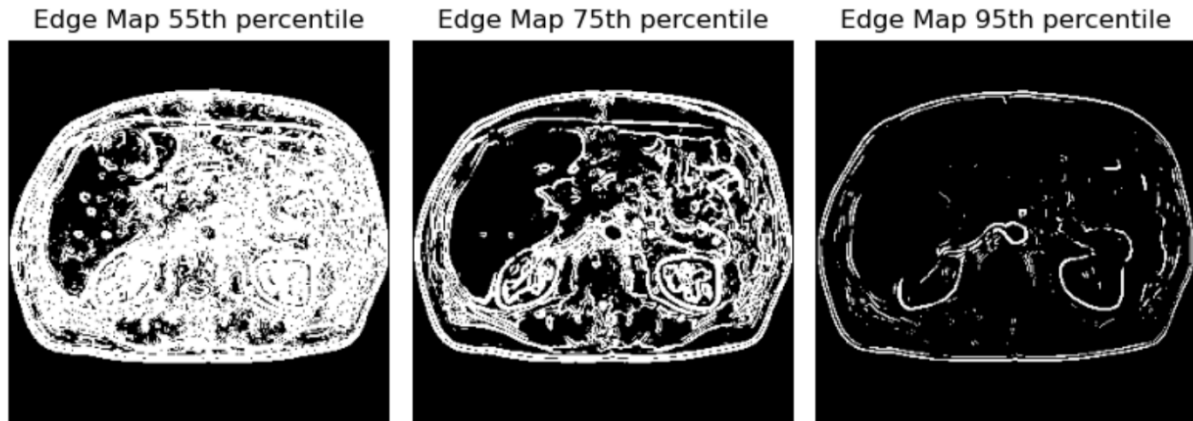


Figure 21. Edge map with different percentiles, 55th percentile (left), 75th percentile (center) and 95th percentile (right).

Figure 22 shows the Sobel edge, Prewitt edge, Canny edge and Laplacian of Gaussian detection algorithms.

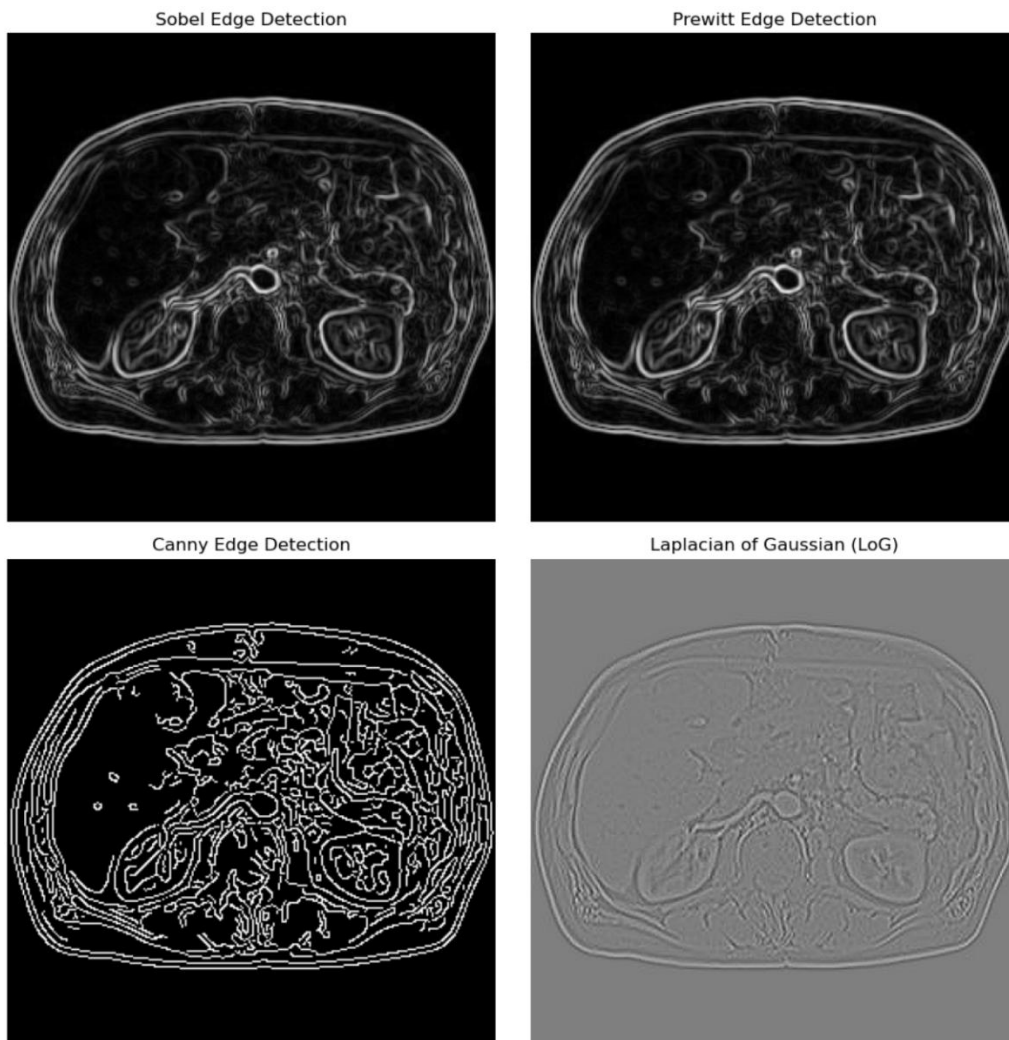


Figure 22. Edge detection results for original image, Sobel edge detection (top left), Prewitt edge detection (top right), Canny edge detection (bottom left), Laplacian of Gaussian (bottom right).

Overall, Canny and Prewitt edge detection produce the most accurate results, with LoG showing similar characteristics with additional detail (but increased complexity).

Question: Compare and discuss the behavior of each operator in the presence of noise. [100 to 150 words]

To evaluate the robustness of edge detection methods, Gaussian noise with a mean of 0 and a standard deviation of 25 was applied to the image, this can be observed in Figure 23.

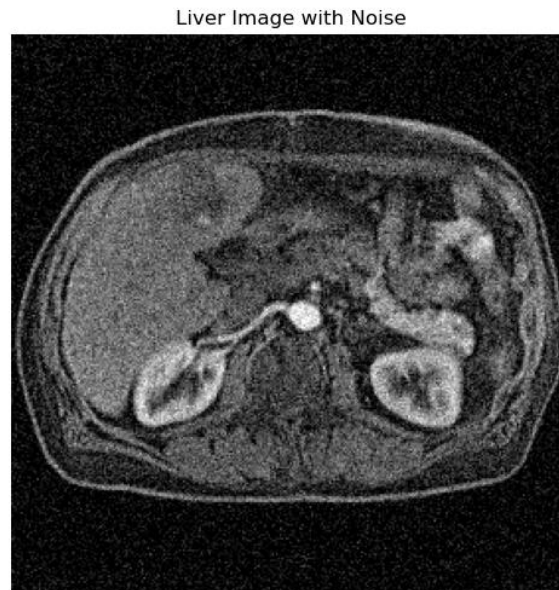


Figure 23. Gray level image with noise

The kernel-based edge detection results become noisy, but the edge map remains somewhat effective in highlighting key structures as can be seen in Figure 24.

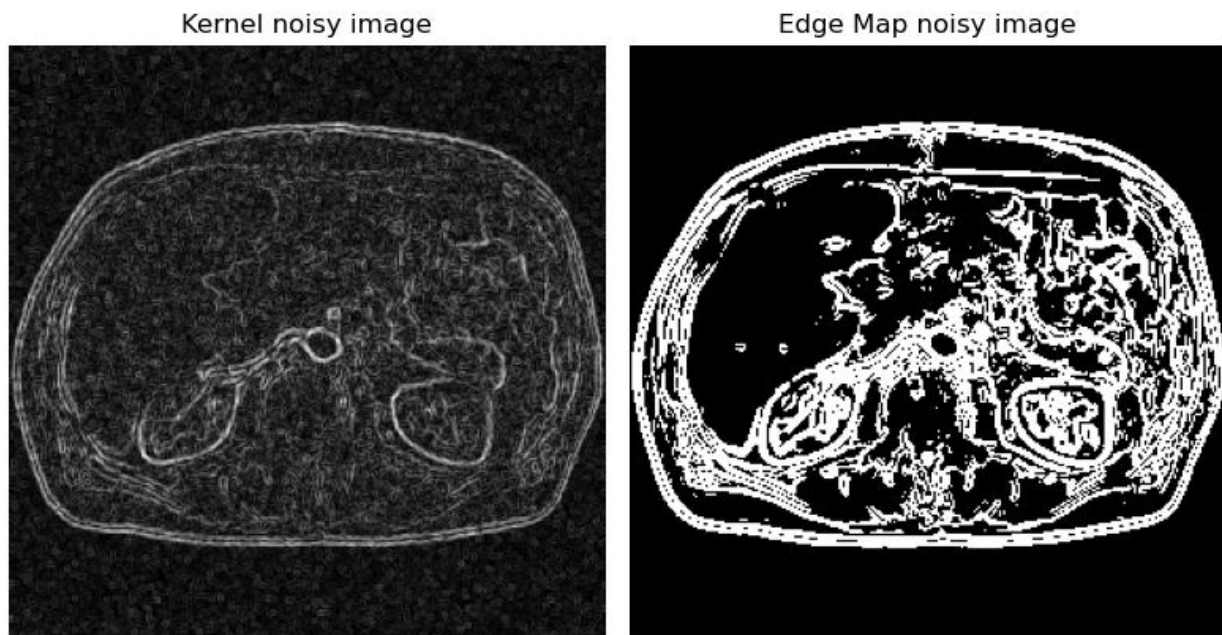


Figure 24. Kernel size 3 for gradient (left) and Edge map with 75th percentile (right).

The presence of Gaussian noise significantly impacts the performance of edge detection operators as can be observed in Figure 25. The Sobel and Prewitt operators rely on gradient calculations and are highly sensitive to noise, this leads to false edges being marked where no actual boundaries exist. The Laplacian of Gaussian (LoG) method performs better in noisy conditions, most likely because it first smooths the image before detecting edges. It effectively reduces false boundaries but loses finer texture details. The Canny edge method is also heavily affected by noise. The gradient computation step amplifies noise, which results in additional edges being marked and a less accurate result.

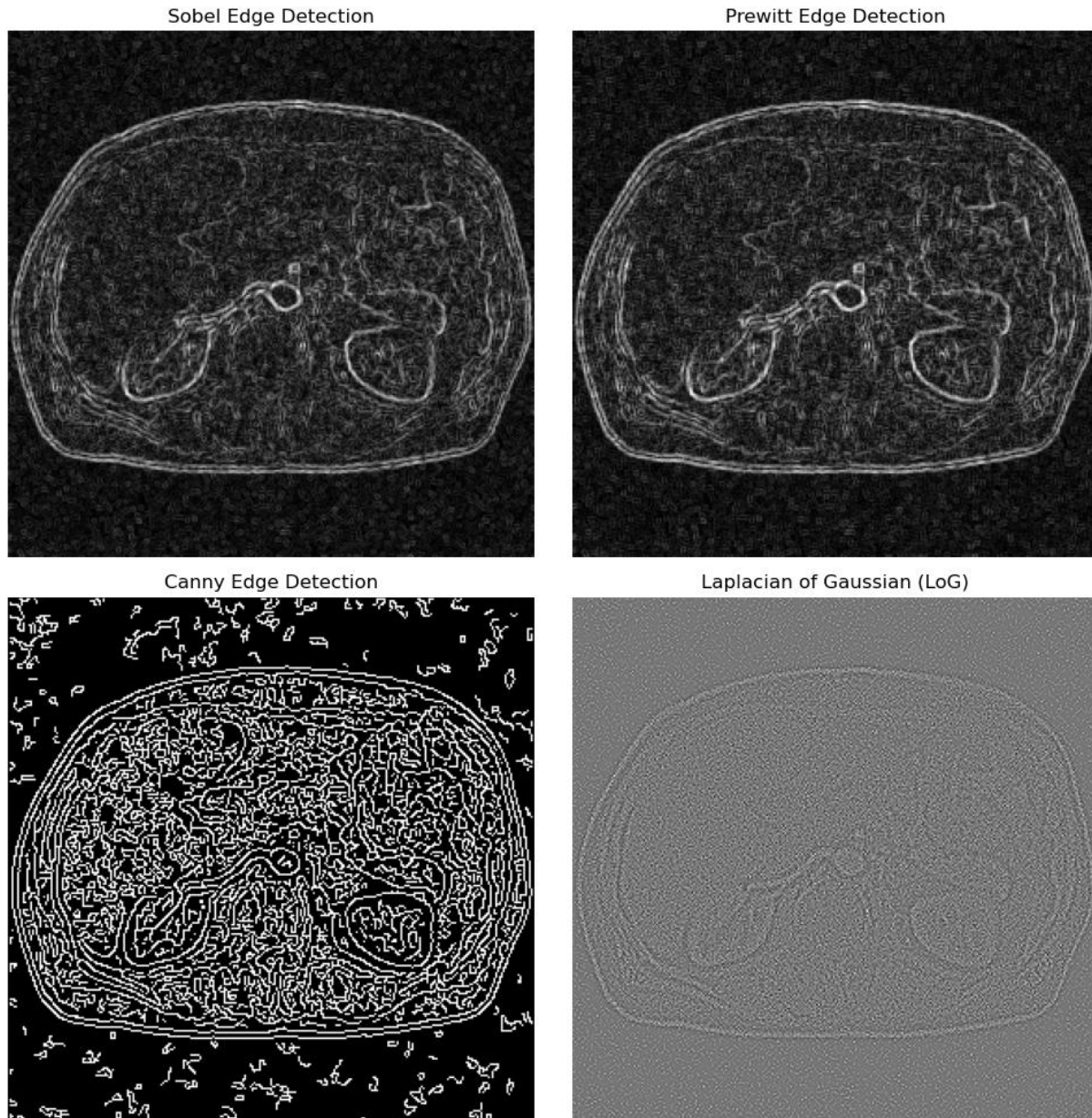


Figure 25. Edge detection results in the image with Gaussian noise. Sobel edge detection (top left), Prewitt edge detection (top right), Canny edge detection (bottom left), Laplacian of Gaussian (bottom right).

References:

- [1] R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, 3rd ed. Upper Saddle River, NJ, USA: Pearson Prentice Hall, 2008.
- [2] E. Ukwatta, *Image Enhancement*, ENGG*6302 Image Processing, Winter 2025, School of Engineering, University of Guelph.
- [3] OpenCV, "OpenCV Documentation," Feb. 2025. Available online: [OpenCV](#) [Accessed: 05-02-25].
- [4] Matplotlib, "Matplotlib Documentation," Feb. 2025. Available online: [Matplotlib](#) [Accessed: 05-02-25].

Declaration: Some parts of the document were enhanced using large language models for better readability.